

Developing an Interactive VOICE RESPONSE SYSTEM

This article helps readers in developing automated speech recognition (ASR) and text-to-speech technology (TTS). This technology assists in automated dissemination of information, taking data input and feedback, authenticating users, etc, remotely through a medium like a mobile phone or a landline.



In India, particularly in areas where teledensity is around 70 per cent and the literacy rate is quite low, voice can become an effective way of delivering services and information to the masses. Such a voice-enabled platform is backed by ASR (automated speech recognition) and TTS (text-to-speech) technology.

Traditionally, an IVR (interactive voice response) system is built on a DTMF (dual tone multi frequency) base input given by users. Whenever users call a customer care number, they hear an audio prompt and give inputs by pressing a number. This is known as the DTMF base input provided by users. But this has very limited functionality. Today, only 22 per cent of the people in the world have access to the Internet. DTMF based IVR systems also limit functionality. For example, AIIMS (All India Institute of Medical Science), New Delhi, has very long queues for the patient appointment system, often leading to hours of delay. AIIMS also has a Web portal for the patient appointment system but most of the patients are not IT literate, though they have access to telephones. So what do they do? They call the number 01126589999. After dialling, the system plays an audio prompt, asking callers to select Hindi or English

by pressing a button. After this, the system asks the caller to enter the UHID number (old patients have a UHID card). When they enter this either via DTMF press or by speech, the system asks the caller: "Which department do you want an appointment for? Say the department's name." On receiving the reply, the system gives a time and date for the patient's appointment. The patient also gets an SMS confirmation. This saves the organisation manpower in fixing the time for an appointment and increases organisational efficiency. Because AIIMS has more than 30 departments, users are asked to speak out the department's name. Let's now learn how to develop a speech based IVR system.

Hardware required for developing speech based IVR

- PRI (primary rate interface) line:** This is a line provided by telephone companies. Using the PRI line, we can get 30 calls per second, both incoming and outgoing, simultaneously. In India, PRI is delivered through the E1 Channel medium, which has a bandwidth of 2.048Mbit/s and 32 channels.
- Cisco Router 2850 and above:** In this router, the IOS (Internet operating system) is 'c2800nm-adventerprisek9-mz.151-4.